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I've been working on this research paper for several months now. What are its fatal weaknesses ? how can i complete it into the platonic ideal shape it yearns to be, guided by its own inner music? what are the top 1000 literal line item fixes I have to make, that altogether UPGRADE the mathematics and the physics? never downgrade? Only tolerate realizing the strongest possible version of this paper imaginable/conceivable.

I need to get this to a publishable shape soon, some time this year (can be end of year) so I can put the preprint on arxiv. Recall that every line of this manuscript needs to be such that one of witten,

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I'll treat this as a referee-style mathematical audit of the manuscript's theorem spine: isolate unsupported claims, type/category mismatches, circular dependencies, and replace them with concrete lemmas, definitions, propositions, and formulas.

The strongest honest version is narrower: many claims must be confined to Koszul/completed/coderived or generic PBW settings, with conditional log-FM and conjectural bridges promoted to explicit firewalls.

Thought for 11m 26s >

According to the uploaded manuscript from 2026-06-05, the paper's **fatal weakness is not lack of ambition**; it is that the strongest ideas are still distributed across conditional lanes, firewalls, and comparison surfaces rather than compressed into one type-correct theorem machine. The manuscript itself identifies the right spine: A, B, D, H, and the proved C0/C1 part of C are the structural skeleton, while shifted-symplectic C2, universal holography, and Calabi–Yau source functors are separate comparison layers used only where their hypotheses are stated. main main

The ideal publishable form is therefore not “more vast”; it is **more exact**. The paper should become a theorem about a signed logarithmic Fulton–MacPherson modular cooperad, an ordered chiral bar construction, a typed Verdier–Koszul duality, and a completed bar–cobar/Positselski equivalence. The strongest version is the one that never lets an object cross a type boundary without a named functor, never lets a raw direct sum stand where a completed inverse limit is required, and never calls a recognition target a constructed object.

The fatal weaknesses

1. The log-FM cooperad is the hidden keystone, but it is still conditional.

The manuscript says Mok supplies strata, codimensions, and degeneration formulae, while the Gysin residue maps, orientation lines, proper pushforwards, automorphism normalisations, and homotopy-coherent coassociativity are additional manuscript data; every $D^2 = 0$ statement using the planted-forest differential d_{pf} is conditional on this signed residue-pushforward package. main This is the first fatal gap: without this cooperad, the modular MC equation is not a theorem; it is a schema.

2. The five-object firewall must become the grammar of every theorem.

The paper correctly distinguishes

$$A, \quad B_X(A), \quad A^i, \quad A^!, \quad Z_{ch}^{der}(A),$$

and says the counit $\Omega_X B_X(A) \rightarrow A$ is bar–cobar inversion, the passage $A^i \mapsto A^!$ is Verdier/Koszul dualization, and the derived centre is neither. main This firewall is currently a remark-level correction. It needs to be the global type system. Every theorem statement should have source, target, ambient category, completion, and functor.

3. Theorem B cannot be stated raw-unconditionally.

The manuscript explicitly says Theorem B has no raw-level unconditional reading: finite-stage statements apply to finite-window conilpotent chiral CDG-coalgebras, completed statements require a strict Mittag-Leffler tower satisfying CP1–CP3, and the raw direct-sum equivalence is genuinely false for class M.

^{main} Any place in the manuscript that still rhetorically treats Theorem B as raw-level universal is mathematically fatal.

4. Class M is not a minor boundary; it is the test of the whole completion formalism.

The manuscript states the raw class-M failure is detected by infinite OPE-mode and harmonic correction families, with the nonzero Virasoro shadow invariant $S_4(\text{Vir}_c) \neq 0$ at generic c . ^{main} The ideal paper must make class M the reason the completed theory exists, not an exception patched later.

5. H_{Δ_5} is a recognition target unless D1–D5 are proved.

The manuscript states that the Δ_5 bar–cobar equivalence depends on compact Hall source, named K3 source datum, parity/root fixtures, PBW/no-extra condition, and completion/comparison maps; without D1–D5 the displayed equivalence is a target, not an unconditional theorem. ^{main} If H_{Δ_5} remains central, D1–D5 must be promoted into a self-contained construction theorem or quarantined as a conditional appendix.

6. Theorem H is not a formal consequence of bar–cobar inversion.

The manuscript correctly states Theorem H requires PBW chiral Koszulness, E_∞ -chiral completion, finite-type/perfectness, genericity, strict Mittag–Leffler completion, and the PBW/Arnold contraction; only then

$$CH^n(A) = 0 \quad (n \notin \{0, 1, 2\})$$

and

$$P_A(t, q) = HS_{Z(A)}(q) + HS_{CH^1(A)}(q)t + HS_{Z(A^1)}(q)t^2.$$

^{main} Any wider claim must exhibit the obstruction complex $KD_H^\bullet(A)$ and prove it acyclic.

7. The paper’s greatest theorem is currently hidden by too many comparative universes.

Volume I should prove the ordered/symmetric modular Koszul theorem on curves. $\text{SC}_{\text{ch}, \text{top}}$, E_3 , CY-to-chiral, K3 Hall, open/closed holography, and arithmetic shadows should appear only after a typed comparison theorem. The paper itself says these are separate layers, not replacements for the bar–cobar spine. ^{main}

The Platonic theorem the paper wants

The strongest central theorem should look like this.

Let X be a smooth algebraic curve over $\mathbb{C}$, possibly inside a simple-normal-crossing pair (X, D) . Let A be an augmented, holonomic, conformal-weight filtered E_1 -chiral algebra over X , finite type in each weight window, equipped with PBW filtration and with all completions strict Mittag–Leffler. Assume a signed logarithmic FM chain cooperad

$$C_{\text{mod}}^{\log FM}$$

has been constructed from Mok strata by graphwise cocompositions

$$\Delta^{\log}(\Gamma) = (\nu(\Gamma))_{\circ} \circ i^{\log}(\Gamma)$$

with determinant-line signs, automorphism normalisations, Gysin residues, proper pushforwards, and homotopy-coherent coassociativity.

Then:

$$B_X^{\text{ord}}(A) = T^c(s^{-1}\bar{A}) \otimes \Omega_{\log}^\bullet(FM_\bullet(X))$$

with

$$d_B = d_A + d_{\text{dR}} + d_{\text{res}}$$

is a complete conilpotent dg coalgebra over $(\text{Ass}^{\text{ch}})^!$, and $d_B^2 = 0$ is equivalent to the Borchers identity transported through the Arnold–Orlik–Solomon relations.

The modular logarithmic enhancement

$$B_X^{\text{mod}, \log}(A)$$

is a coalgebra over $C_{\text{mod}}^{\log FM}$, and its total differential

$$D_{\text{mod}}^{\log} = d_{\dot{C}} + d_{\text{Ch}_{\infty}} + d_{\text{coll}} + d_{\text{sew}} + d_{\text{pf}} + \hbar \Delta$$

satisfies

$$(D_{\text{mod}}^{\log})^2 = 0.$$

Equivalently, the universal class

$$\Theta_A \in \text{Conv}_{\infty}(C_{\text{mod}}^{\log FM}, \text{End}_{\text{Ch}_{\infty}(A)})$$

satisfies

$$D\Theta_A + \frac{1}{2}[\Theta_A, \Theta_A] = 0.$$

The averaging map

$$\text{av}: B_X^{\text{ord}}(A) \rightarrow B_X^{\Sigma}(A)$$

is a morphism only under the stated finite-type or completed equivariance assumptions. It sends the ordered residue $r_A(z)$ to the scalar characteristic $\kappa(A)$, but it is not bar–cobar inversion, not Verdier duality, and not the derived centre.

On the PBW–Koszul locus:

$$\Omega_X B_X(A) \sim A.$$

The Koszul dual is instead

$$A^! := \Omega_X \mathbb{D}_{\text{Ran}} B_X(A)$$

whenever the Verdier–dual completed cobar construction converges.

For class M, the raw direct-sum statement is false; the valid statement is the weight-completed coderived/contraderived Positselski equivalence on strict Mittag–Leffler towers satisfying CP1–CP3.

Finally, under Theorem H hypotheses,

$$CH_{\text{ch}}^n(A) = 0 \quad (n \notin \{0, 1, 2\}),$$

and off that locus the defect is measured by the collision-depth obstruction complex

$$KD_H^{\bullet}(A).$$

Everything else—holography, E_3 , CY source, K3 Hall, arithmetic—must be a typed comparison theorem into this core.

Top 1000 literal line-item mathematical fixes

A. Core theorem architecture: 1–80

1. State one main theorem named **The Modular Ordered Bar Theorem** before all examples.
2. State a separate theorem named **Signed Log-FM Cooperad Theorem**.
3. State a separate theorem named **Typed Verdier–Koszul Firewall Theorem**.
4. State a separate theorem named **Completed Chiral Positselski Theorem**.
5. State a separate theorem named **Curve-Level Chiral Hochschild Amplitude Theorem**.
6. Replace “five theorem packages” by one theorem with five clauses and typed hypotheses.
7. Put $A, B_X(A), A^i, A^!, Z_{\text{ch}}^{\text{der}}(A)$ in every major theorem header.
8. Define the ambient category of A as $\text{Alg}_{\text{Ass}^{\text{ch}}}(\mathcal{D}_X)$.
9. Define the ambient category of $B_X(A)$ as complete conilpotent chiral coalgebras.
10. Define the ambient category of A^i as homology coalgebras of the bar construction.

11. Define the ambient category of $A^!$ as post-Verdier completed chiral algebras.
12. Define the ambient category of $Z_{\text{ch}}^{\text{der}}(A)$ as chiral Hochschild cochains.
13. Remove all informal uses of “bulk” before an open/closed comparison map is constructed.
14. Replace every “is” between different firewalled objects by a named functor.
15. Replace every “recovers the dual” after $\Omega B(A)$ by “recovers A ”.
16. Define $\text{Kosz}(X)$ as a full subcategory with explicit PBW and completion conditions.
17. Define $\text{Kosz}_{E_1}^{\text{ord}}(X)$ separately from $\text{Kosz}_{E_\infty}^\Sigma(X)$.
18. State that $B^{\text{ord}} \rightarrow B^\Sigma$ requires equivariant descent.
19. State that $B^\Sigma(A)$ need not exist for genuinely E_1 -chiral inputs.
20. Insert a diagram:

$$B^{\text{ord}}(A) \rightarrow B^\Sigma(A), \quad \Omega B(A) \rightarrow A, \quad \mathbb{D}B(A) \rightarrow A^!, \quad C_{\text{ch}}^\bullet(A, A) \rightarrow Z_{\text{ch}}^{\text{der}}(A).$$

21. Prove the diagram has no unlabelled commuting arrows.
22. Put all theorem hypotheses in “finite”, “completed”, and “raw” columns.
23. Use “raw” only for direct sums with discrete topology.
24. Use “completed” only for strict inverse limits.
25. Use “finite window” only for finite-dimensional conformal-weight truncations.
26. Define strict Mittag–Leffler once and reference it everywhere.
27. Define all completions by universal properties, not by intuition.
28. Replace $\oplus$ notation by explicit $\lim_N \oplus_{w \leq N}$.
29. State when Hom means continuous Hom .
30. State when tensor products are completed.
31. State when duals are Verdier duals.
32. State when duals are continuous linear duals.
33. State when duals are finite arity duals.
34. Remove all unqualified “dual” language.
35. Replace “Koszul dual” by “Verdier–Koszul dual” when $\mathbb{D}_{\text{Ran}}$ is used.
36. Replace “bar dual” by “bar-homology coalgebra” when referring to A^i .
37. Replace “centre” by $Z_{\text{ch}}^{\text{der}}(A)$, $Z(A)$, or categorical centre explicitly.
38. Define $Z(A) = CH_{\text{ch}}^0(A)$ only on the curve-level chiral Hochschild complex.
39. Never identify $Z_{\text{ch}}^{\text{der}}(A)$ with a physical bulk without OCA.
40. Add theorem: OCA is a quasi-isomorphism iff specified open/closed factorization comparison holds.
41. Separate genus-zero theorems from all-genera theorems.
42. Separate smooth-curve theorems from log-relative theorems.
43. Separate ordered theorems from symmetric coinvariant theorems.
44. Separate chain-level quasi-isomorphism from derived equivalence.
45. Separate coderived equivalence from strict chain homotopy equivalence.
46. Separate chiral Hochschild from algebraic Hochschild.
47. Separate chiral Hochschild from topological Hochschild.
48. Separate curve-level Hochschild from CY categorical Hochschild.
49. Define “uniform-weight lane” mathematically as a diagonal subcomplex.
50. Define “cross-channel correction” as complement of that diagonal projection.
51. Replace “shadow” by a projection map with source and target.
52. Define the shadow tower as a filtered associated graded of Θ_A .
53. Define $S_r(A)$ as an actual cohomology class.
54. Define the obstruction differential that makes $S_r(A)$ closed.
55. Define equivalence relation on shadow towers.
56. Define when two shadow towers classify the same algebra.
57. State fingerprint completeness only for the standard landscape.
58. Remove all global classification rhetoric outside that landscape.
59. Define G/L/C/M as strata of an invariant map, not adjectives.
60. Define r_{max} as supremum of nonzero primitive obstruction degree.
61. Define $\kappa(A)$ as a map H^0 of the degree-two residue complex.
62. Define $K_\kappa(A, A^!)$ only after $A^!$ exists.
63. Define $K_c(A, A^!)$ separately.
64. Define $K_{\text{ghost}}^{\text{leg}}(A)$ as BRST-presentation dependent.

65. Add a proposition proving no equality among $K_c, K_\kappa, K_{\text{ghost}}$ is formal.
66. For each equality among scalar packages, give a witness theorem.
67. Remove every equality of scalar packages without proof.
68. Replace “universal conductor” by a morphism of filtered complexes.
69. Prove the Reynolds projector is continuous in each completed arity.
70. Prove averaging commutes with d_B only under equivariance.
71. State characteristic zero is required for $1/n!$.
72. State what replaces averaging in R -twisted descent.
73. Define the R -twisted coinvariant complex explicitly.
74. Add a lemma that R -twisted coinvariants preserve the bar differential.
75. Add a lemma that R -twisted coinvariants preserve the convolution bracket.
76. State when averaging is a dg Lie morphism.
77. State when averaging is only a linear projection.
78. Remove all uses of “sole information loss” unless the kernel is characterized.
79. Add theorem identifying $\ker(\text{av})$ in degree two as ordered r -matrix data.
80. Add theorem identifying higher $\ker(\text{av})$ pieces as braid/associator data.

B. Logarithmic Fulton–MacPherson cooperad: 81–170

81. Define $FM_n(X|D)$ with its log structure.
82. Define the planted-forest indexing category $\text{For}^{\text{plant}}(n)$.
83. Define morphisms in $\text{For}^{\text{plant}}(n)$ as contractions/graftings.
84. Define orientation line $\text{or}(\tau)$ for each planted-forest type τ .
85. Define the chain group $C_\tau := C_*(S_\tau; \text{or}(\tau))$.
86. Define the differential as signed sum over codimension-one face maps.
87. Prove $d^2 = 0$ for the planted-forest face complex.
88. Define Mok crossing number $c(\tau)$.
89. Define pure forest depth $p(\tau)$.
90. Define separating-node count $s(\Gamma)$.
91. Define non-separating-node count $\ell(\Gamma)$.
92. Define bar-contraction count $b(\Gamma)$.
93. Define refined degree $\mu = (g, r; b, s, \ell, c, p)$.
94. Prove $\text{codim}_{\log} = s + \ell + c + p$.
95. Prove old tridegree is the projection $(g, r, c + p)$.
96. State when b contributes to homological degree but not log codimension.
97. Define ν_Γ in $\Delta^{\log}(\Gamma) = (\nu_\Gamma)_* \Delta^{\log}(\Gamma)$.
98. Define $i_\Gamma^!$ as a Gysin/residue map.
99. Prove $i_\Gamma^!$ is functorial under nested strata.
100. Prove excess-intersection correction vanishes or specify it.
101. Define proper pushforward $(\nu_\Gamma)_*$ in the chosen chain category.
102. Prove $(\nu_\Gamma)_*$ respects orientation twists.
103. Compute automorphism normalisation for a graph with repeated identical vertices.
104. Prove automorphism factors make cocomposition invariant under graph isomorphism.
105. Define $\Delta^{\log}(\Gamma)$ on generators.
106. Prove $\Delta^{\log}(\Gamma_1 \circ \Gamma_2) = (\Delta^{\log}(\Gamma_1) \otimes 1) \Delta^{\log}(\Gamma_2)$.
107. Prove coassociativity up to specified homotopy.
108. If only homotopy-coassociative, define the full A_∞ -cooperad structure.
109. Prove the homotopies satisfy pentagon coherence.
110. Define determinant-line sign for two edge contractions.
111. Prove swapping contraction order multiplies by -1 .
112. Prove codimension-two boundary cancellations.
113. Compute the binary collision boundary explicitly.
114. Compute the triple collision boundary explicitly.
115. Compute the first planted-forest/Mok crossing boundary explicitly.
116. Compute a mixed stable-node/forest boundary explicitly.
117. Compute a separating/non-separating mixed boundary explicitly.

118. Add a theorem that $C_{\text{mod}}^{\log FM}$ is a modular cooperad.
119. State whether it is dg, curved dg, or homotopy dg.
120. Define its curvature if curved.
121. Prove curvature vanishes after including d_{pf} .
122. Define the smooth-curve subcooperad C_{sm}^{FM} .
123. Prove $C_{\text{sm}}^{FM} \hookrightarrow C_{\text{mod}}^{\log FM}$.
124. Define the associated graded for the log filtration.
125. Prove

$$\text{gr}^F \Omega B^{\text{mod}, \log}(A) \cong \Omega(\text{gr}^F B^{\text{mod}, \log}(A)).$$

126. State exact boundedness hypotheses for the associated-graded isomorphism.
127. Prove completions commute with gr^F .
128. Prove residues commute with gr^F .
129. Define the d_{pf} term as an operator of bidegree $(0, 1)$ or specify its degree.
130. Prove d_{pf} anticommutes with d_{coll} .
131. Prove d_{pf} anticommutes with d_{sew} .
132. Prove d_{pf} anticommutes with $\hbar \Delta$.
133. Prove d_{pf} anticommutes with Čech differential.
134. Prove d_{pf} anticommutes with transferred Ch_{∞} differential.
135. Compute $(d_{\text{pf}})^2$ explicitly.
136. State the sign convention making $(d_{\text{pf}})^2 = 0$.
137. Define all six components of $D_{\text{mod}}^{\log}$ in one table.
138. Give their cohomological degrees.
139. Give their filtration degrees.
140. Give their graph-theoretic effects.
141. Give their arity effects.
142. Give their genus effects.
143. Give their conformal-weight effects.
144. Prove each pairwise anticommutator.
145. Add a master $D^2 = 0$ proof as one theorem.
146. Remove any $D^2 = 0$ claim preceding that theorem.
147. Define the relative universal stable family used.
148. Define base stack and boundary divisors.
149. State whether stacks are orbifolds, DM stacks, or derived stacks.
150. Define chains on stacks with automorphism groups.
151. Prove pushforward is compatible with stack stabilizers.
152. Prove coinvariants over automorphism groups are finite-dimensional in each window.
153. Define local coordinates near log boundary.
154. Compute a local residue in those coordinates.
155. Compare local residue with Gysin $i^!$.
156. Prove residue independence of coordinate choice.
157. Add a lemma on normal-crossing compatibility.
158. Add a lemma on nested blow-up order independence.
159. Prove planted forests encode nested diagonals injectively.
160. Prove every relevant boundary stratum is represented.
161. Prove no extra strata occur.
162. Define tropicalisation of log-FM spaces.
163. Prove face poset equals planted-forest poset.
164. State exact relation to ordinary FM when $D = \emptyset$.
165. Prove log-FM reduces to ordinary FM in that case.
166. Define "Mok crossings" by a universal property.
167. Prove their count is stable under base change.
168. Prove degeneration formula is compatible with chiral operations.
169. Separate Mok's theorem from manuscript-added chain data.
170. Add final corollary: modular convolution L_{∞} -algebra is well-defined only after 81–169.

C. Ordered chiral bar construction: 171–250

171. Define $B_X^{\text{ord}}(A) = T^c(s^{-1}\bar{A})$ in $\mathcal{D}_X$ -modules.
172. Define whether tensor powers are external, chiral, or factorization tensor powers.
173. Define the coefficient complex on $FM_n(X)$.
174. Define the full term $B_n(A)$ including log forms.
175. Define d_A on desuspended tensors.
176. Define d_{dR} on logarithmic forms.
177. Define d_{res} using OPE modes.
178. State exact sign of d_{res} .
179. Compute d_{res} on $s^{-1}a \otimes s^{-1}b$.
180. Compute d_{res} on three tensors.
181. Prove $d_A^2 = 0$.
182. Prove $d_{\text{dR}}^2 = 0$.
183. Prove $d_A d_{\text{dR}} + d_{\text{dR}} d_A = 0$.
184. Prove $d_A d_{\text{res}} + d_{\text{res}} d_A = 0$ from A -differential compatibility.
185. Prove $d_{\text{dR}} d_{\text{res}} + d_{\text{res}} d_{\text{dR}} = 0$ by Stokes/residue compatibility.
186. Prove $d_{\text{res}}^2 = 0$ by Borchers plus Arnold.
187. State Arnold relation in your exact sign convention.
188. State Borchers identity in your exact grading convention.
189. Prove the three-term Arnold identity implies Jacobi for $a_{(0)}b$.
190. Prove higher Borchers terms match higher residues.
191. Define which OPE modes are seen by $\eta_{ij} = d \log(z_i - z_j)$.
192. Explain how second-order poles enter after residue contraction.
193. Define $r_A(z)$ as a class in $A \otimes A \otimes \Omega^1$, not a scalar.
194. Prove $r_A(z)$ solves CYBE when the OPE bracket satisfies Jacobi.
195. Prove scalar $\kappa(A)$ is the averaged degree-two trace of $r_A(z)$.
196. Define deconcatenation coproduct with Koszul signs.
197. Prove deconcatenation is coassociative.
198. Prove d_B is a coderivation.
199. Prove $B_X^{\text{ord}}(A)$ is conilpotent by bar length.
200. Define the completion when bar length is infinite.
201. Prove completed conilpotence in each finite weight window.
202. Define augmentation $\epsilon : A \rightarrow \omega_X$.
203. Define $\bar{A} = \ker \epsilon$.
204. Prove augmentation ideal is preserved by OPE differential.
205. Define unit/counit in the bar coalgebra.
206. Prove reduced bar removes vacuum degeneracies.
207. Define Ass^{ch} .
208. Define $(Ass^{\text{ch}})^!$.
209. Prove $B_X^{\text{ord}}(A)$ is a coalgebra over $(Ass^{\text{ch}})^!$.
210. State finite-local-free assumptions for Ass^{ch} self-duality.
211. Prove Ass^{ch} quadratic dual is sign-twisted itself.
212. Define ordered Ran space $Ran^{\text{ord}}(X)$.
213. Prove the bar construction is a factorization coalgebra on $Ran^{\text{ord}}(X)$.
214. Define restriction maps for disjoint opens.
215. Prove cosheaf/factorization property for B^{ord} .
216. Prove compatibility of residues with factorization product.
217. Define ordered FM compactification.
218. Prove it maps to unordered FM by quotient.
219. Define symmetric bar $B^{\Sigma}(A)$.
220. Prove $B^{\Sigma}(A)$ exists only under locality/descent.
221. Prove $B^{\text{ord}} \rightarrow B^{\Sigma}$ is a morphism under descent.
222. Compute kernel in arity two.
223. Compute kernel in arity three.
224. Relate arity-three kernel to associator class.
225. Prove averaging kills antisymmetric residue classes.

- 226. Prove averaging preserves scalar κ .
- 227. Define KZ pullback identity $d_B = KZ^*(\nabla_{\text{Arnold}})$ precisely.
- 228. Specify the representation of the infinitesimal braid Lie algebra.
- 229. Prove $KZ^*(\nabla_{\text{Arnold}})$ acts on B^{ord} .
- 230. Prove flatness iff Arnold plus infinitesimal braid relations.
- 231. State when this identity holds only genus zero.
- 232. Define elliptic correction at genus one separately.
- 233. Define higher-genus prime-form correction separately.
- 234. Remove global $d_B = KZ^*(\nabla)$ unless genus and connection are specified.
- 235. Add local coordinate independence theorem for d_B .
- 236. Prove coordinate change gives Virasoro cocycle.
- 237. State when stress tensor is required.
- 238. Prove d_B is natural under étale maps of curves.
- 239. Prove d_B is natural under smooth base change with holonomic finiteness.
- 240. Add comparison with BD chiral bracket as a theorem.
- 241. Prove OPE residue equals BD chiral operation.
- 242. Prove no higher poles are lost in the bar construction.
- 243. Give explicit formula for an arbitrary n -th product contribution.
- 244. State all signs in Appendix B once and reference them.
- 245. Remove duplicate sign appendices or make them non-redundant.
- 246. Add a "bar sign theorem" proving all displayed conventions agree.
- 247. Compute d_B^2 in $n = 3$ fully.
- 248. Compute d_B^2 in $n = 4$ through one nested boundary.
- 249. State the general n induction.
- 250. End section with theorem: B_X^{ord} is a functor from augmented E_1 -chiral algebras to complete conilpotent dg coalgebras.

D. Bar–cobar and Verdier–Koszul duality: 251–330

- 251. Define Ω_X on complete conilpotent chiral coalgebras.
- 252. Define its underlying completed free chiral algebra.
- 253. Define the cobar differential.
- 254. State its signs.
- 255. Prove $\Omega_X \dashv B_X$.
- 256. Identify unit and counit explicitly.
- 257. Prove the counit $\Omega_X B_X(A) \rightarrow A$ is natural.
- 258. Prove the unit $C \rightarrow B_X \Omega_X(C)$ is natural.
- 259. State when the adjunction is Quillen.
- 260. Define weak equivalences detected by bar–cobar counit.
- 261. Define Koszul locus by counit quasi-isomorphism.
- 262. Prove equivalence with PBW E_2 -collapse.
- 263. Prove equivalence with diagonal Ext vanishing.
- 264. Prove equivalence with twisted tensor acyclicity.
- 265. Prove equivalence with FM boundary acyclicity.
- 266. Keep factorization homology criterion conditional.
- 267. Keep D-module purity implication one-way unless converse proved.
- 268. State Lagrangian criterion as conditional on perfectness/nondegeneracy.
- 269. Define $A^i = H^\bullet(B_X(A))$ as a coalgebra.
- 270. Prove A^i lies on Koszul diagonal under PBW.
- 271. Define $\mathbb{D}_{\text{Ran}} B_X(A)$ only for holonomic dualizable objects.
- 272. Prove $B_X(A)$ is holonomic under stated assumptions.
- 273. Prove Verdier dual of coalgebra is algebra.
- 274. State tensor exchange $!$ -tensor versus $*$ -tensor.
- 275. Define $A_\infty^! := \Omega_X \mathbb{D}_{\text{Ran}} B_X(A)$.
- 276. Define strict $A^!$ as formal/minimal model of $A_\infty^!$.
- 277. Prove $H^\bullet(A_\infty^!) \cong (A^i)^\vee$ in finite type.
- 278. State when $A_\infty^!$ is formal.

279. Do not call $A^!$ strict unless formality is proved.
280. Add theorem: $\Omega B(A) \not\cong A^!$ unless special self-dual theorem.
281. Add examples: Heisenberg.
282. Add examples: affine Kac–Moody.
283. Add examples: Virasoro.
284. Add examples: W_3 .
285. For each example, list $A, B(A), A^i, A^!, Z^{\text{der}}$.
286. Prove level reflection $k \mapsto -k - 2h^\vee$ only in affine lane.
287. Do not transfer level reflection to $\bar{W}$ -algebras without DS proof.
288. Define DS reduction as a functor on filtered chiral algebras.
289. Prove DS commutes with bar construction under hypotheses.
290. Prove DS commutes with Verdier duality under hypotheses.
291. Define the DS-transferred Koszul dual.
292. State obstructions to DS/bar commutation.
293. Define non-quadratic Koszul duality via curved A_∞ completion.
294. Prove convergence of curved cobar for each non-quadratic family.
295. Define m_0 curvature.
296. Prove $m_1^2 = [m_0, -]$.
297. State centrality of m_0 .
298. Prove centrality in Heisenberg.
299. Prove centrality in affine generic lane.
300. State failure at critical level.
301. Define coderived category $D_{\text{ch}}^{\text{co}}(C)$.
302. Define contraderived category $D_{\text{ch}}^{\text{ctr}}(C)$.
303. Define chiral CDG-comodule.
304. Define chiral CDG-contramodule.
305. Prove chiral Positselski equivalence finite-stage.
306. Prove it is a lift of classical Positselski under chiral locality.
307. State exact finite-dimensional graded-piece condition.
308. State exact conilpotence condition.
309. State exact curvature condition.
310. Define CP1–CP3.
311. Prove strict ML inverse limit preserves equivalence.
312. Prove continuous homotopies exist in the completed limit.
313. Prove raw direct-sum class M has no such homotopy.
314. Compute Virasoro obstruction element $\xi_N = \sum_{|k| \leq N} s^{-1} L_{-k} \otimes s^{-1} L_k$.
315. Prove ξ_N has no raw direct-sum limit.
316. Prove ξ_N has completed inverse-limit element.
317. Prove same mechanism for W_N higher-spin modes.
318. Prove same mechanism for critical Kac–Moody screening corrections.
319. Replace all raw class-M Theorem B statements by completed ones.
320. Add “false theorem” box: raw class-M Positselski is false.
321. Prove G and L raw cases.
322. Prove finite free-field C raw cases.
323. State integer-spin/infinite free-field C requires completion.
324. State Yangian finite-bar-degree raw theorem separately.
325. State Yangian generic spectral module theorem as completed.
326. State Feigin–Frenkel center theorem as completed.
327. Define completion topology on Feigin–Frenkel center.
328. Prove finite-dimensionality of conformal-weight pieces.
329. Prove CP1–CP3 for each standard family claimed.
330. If CP1–CP3 not proved, mark the theorem conditional.

E. Theorem H and chiral Hochschild: 331–400

331. Define $CH_{\text{ch}}^*(A)$ from diagonal bar resolution.
332. Define $RHH_{\text{ch}}(A)$ as the completed chiral Hochschild complex.

333. Define collision-depth filtration F_p^{FM} .
334. Define $gr_p^{FM} RH H_{ch}(A)$.
335. Define the collision-depth spectral sequence.
336. Define $\mathfrak{o}_{H,1}^{\geq 3}(A)$.
337. Define $\mathfrak{o}_{H,\infty}^{\geq 3}(A)$.
338. Define $KD_H^\bullet(A)$.
339. Prove Theorem H is equivalent to $KD_H^\bullet(A)$ acyclicity in high degrees.
340. State PBW chiral Koszul hypotheses explicitly.
341. State E_∞ -completion hypothesis explicitly.
342. State finite-window perfectness explicitly.
343. State genericity exclusions explicitly.
344. State critical-level exclusion explicitly.
345. Prove positive Arnold fibre contraction.
346. Identify the Shelton–Yuzvinsky contraction used.
347. Prove it transports through chiral quadratic-Koszul comparison.
348. Prove descent from ordered to symmetric Hochschild complex.
349. Prove Mittag–Leffler completion commutes with totalization.
350. Prove $CH^n = 0$ for $n > 2$.
351. Prove curve dimension creates the amplitude ceiling 2.
352. Prove the t^2 -column is $Z(A^1)^\vee \otimes \omega_X$.
353. Define $CH^1(A)$ explicitly as deformation tangent.
354. Compute $CH^\bullet$ for Heisenberg.
355. Compute $CH^\bullet$ for affine $\mathfrak{sl}_2$ generic.
356. Compute $CH^\bullet$ for affine $\mathfrak{sl}_3$ generic.
357. Compute $CH^\bullet$ for Virasoro generic.
358. Compute $CH^\bullet$ for W_3 generic.
359. State which computations are universal algebra and which are simple quotient.
360. Remove all Theorem H claims for logarithmic triplets unless hypotheses proved.
361. State triplet $W(p)$ Koszulness as conjectural.
362. Define the degree-three triplet obstruction $\mathfrak{o}_{H,W}^3$.
363. State C_2 -cofiniteness does not imply chiral Koszulness.
364. State rationality does not imply Theorem H.
365. State finite module category does not imply Theorem H.
366. For admissible affine quotients, define Φ_H^{adm} .
367. Prove Φ_H^{adm} before using admissible obstruction classes.
368. Define Cartan bar obstruction for $\mathfrak{sl}_3$ admissible quotients.
369. Prove whether it injects into full obstruction.
370. For critical affine level, state Theorem H is false or inapplicable.
371. Prove Sugawara denominator obstruction.
372. Prove curvature-residue obstruction $\mathfrak{c}_{crit}(\mathfrak{g})$.
373. Prove Ext diagonal concentration fails at critical level.
374. Separate Feigin–Frenkel center from full Hochschild complex.
375. Remove any implication “finite degree-zero center $\Rightarrow$ amplitude bound.”
376. Define coHochschild transport.
377. Prove bimodule–bicomodule equivalence before coHochschild vanishing.
378. State no coHochschild theorem for arbitrary coalgebra.
379. Prove Hilbert polynomial finite q -coefficients.
380. State $P_A(t, q)$ is polynomial in t , not necessarily q .
381. Add exact sequence for off-Koszul high-degree tail.
382. Define $\partial_H^3(A)$.
383. Prove $CH^3(A) = \text{coker } \partial_H^3(A)$.
384. Prove $CH^n(A) = H^n(KD_H^\bullet(A))$ for $n \geq 4$.
385. Compute $KD_H^\bullet$ for one non-Koszul example.
386. Compute $KD_H^\bullet$ for Virasoro at a singular c .
387. Compute $KD_H^\bullet$ for a triplet algebra if claimed.
388. Define low-column Verdier identification.

- 389. State when low-column identification fails.
- 390. Remove all free-polynomial algebra claims unless Massey vanishing is proved.
- 391. Define Massey products in $CH_{\text{ch}}^*(A)$.
- 392. Prove FM formality implies their vanishing under hypotheses.
- 393. State if FM formality is rational, real, or integral.
- 394. Prove compatibility with conformal-weight completion.
- 395. Add a theorem comparing mode Hochschild to chiral Hochschild only as a filtered map.
- 396. State it is not an equivalence unless proved.
- 397. Add a theorem comparing Lie algebra cohomology to chiral Hochschild only on image.
- 398. State all Hochschild variants in a three-column table.
- 399. Move topological Hochschild remarks out of Theorem H proof.
- 400. End Theorem H section with exact boundary firewall.

F. E_1, E_∞, R -matrices, Yangians: 401–470

- 401. Define E_∞ -chiral as commutative algebra in $(D\text{-mod}(X), \otimes^{ch})$.
- 402. Define E_1 -chiral as algebra over Ass^{ch} .
- 403. State E_1 -chiral on a curve is not topological E_1 .
- 404. Define P_∞ -chiral separately.
- 405. State hierarchy $E_\infty \subset P_\infty \subset E_1$.
- 406. List the five presentations of E_1 -chiral structure.
- 407. Prove none are automatically equivalent.
- 408. State H1–H4 comparison hypotheses.
- 409. Prove strictification Quillen equivalence under H1.
- 410. Prove EK reconstruction Quillen equivalence under H2.
- 411. Prove ordered BD realization under H3.
- 412. Prove preservation of twisting morphisms under H4.
- 413. Remove all cross-presentation transport without H1–H4.
- 414. Treat double- A_∞ presentation as open unless proved.
- 415. Define ordered Hochschild $HH_{\text{ord}}^{E_1}(A, A)$.
- 416. Use it for genuinely E_1 inputs.
- 417. State chiral Deligne–Tamarkin theorem requires E_∞ -chiral input.
- 418. Do not use symmetric bar for Yangians unless descent is supplied.
- 419. Define ordered r -matrix $r_A(z)$ before scalar $\kappa(A)$.
- 420. Prove CYBE from Arnold relation for $r_A(z)$.
- 421. Define modified/quantum YBE for $R(z)$.
- 422. State how r quantizes to R .
- 423. Prove RTT relations in the chosen finite window.
- 424. Define RTT window $A_{\leq N}^{RTT}$.
- 425. Prove PBW for $A_{\leq N}^{RTT}$.
- 426. Prove $R_{\leq N}^\perp = \text{im } \Phi_{R^{-1}}$.
- 427. Prove scalar gauge equivalence $R_h^{-1} \sim R_{-\hbar}$.
- 428. Define spectral Laurent completion.
- 429. State Yangian duality only after 424–428.
- 430. Distinguish Drinfeld second presentation from RTT presentation.
- 431. Prove comparison map between them.
- 432. Do not transfer exceptional RTT claims without finite module packet.
- 433. Define finite-dimensional module packet.
- 434. Prove trace pairing nondegenerate on it.
- 435. Compute $Y_{\hbar}(\mathfrak{sl}_2)$ RTT dual explicitly.
- 436. Compute $Y_{\hbar}(\mathfrak{sl}_3)$ RTT dual explicitly.
- 437. State exceptional types require separate obstruction theorem.
- 438. Define affine Yangian completion.
- 439. Define shifted Yangian completion.
- 440. State which bar coalgebra is conilpotent by bar degree.
- 441. Prove finite-dimensional graded pieces at finite bar degree.
- 442. Define generic spectral-parameter module surface.

- 443. State it is completed, not raw.
- 444. Define Drinfeld–Kohno comparison as a functor between categories.
- 445. Prove chain-level DK only on evaluation locus.
- 446. Define extension beyond evaluation locus as conditional.
- 447. Separate Yangian from Koszul dual of affine $V_k(\mathfrak{g})$.
- 448. Separate Yangian from derived centre.
- 449. Separate $r_{\text{coll}}(z) = k\Omega/z$ from $r_{\text{KZ}}(z) = \Omega/((k + h^\vee)z)$.
- 450. Explain the shift $k + h^\vee$ in KZ normalization.
- 451. Prove Sugawara normal-ordering contribution to κ .
- 452. State critical level $k = -h^\vee$ is excluded for KZ normalization.
- 453. Compute Heisenberg $r(z) = k/z$.
- 454. Compute affine $r(z) = k\Omega/z$.
- 455. Prove averaging sends scalar Heisenberg r to k .
- 456. Prove averaging sends affine r to the declared κ .
- 457. Define trace normalization of Ω .
- 458. State all root-length normalizations.
- 459. Define $h^\vee$ conventions.
- 460. Define k conventions.
- 461. Remove inconsistent $k + h^\vee$ powers.
- 462. Check whether $\kappa(V_k(\mathfrak{g}))$ is linear or quadratic in $k + h^\vee$.
- 463. State one formula and use it everywhere.
- 464. Check $\kappa(\text{Vir}_c) = c/2$ everywhere.
- 465. Check $K_\kappa(\text{Vir}_c, \text{Vir}_{26-c}) = 13$.
- 466. Check $K_c(\text{Vir}) = 26$.
- 467. Check $K_\kappa(W_3) = 250/3$ under your $H_N - 1$ convention.
- 468. Check $K_c(W_3) = 100$.
- 469. Add a scalar-normalization appendix.
- 470. Make all tables depend on that appendix.

G. Genus tower and modular characteristic: 471–540

- 471. Define $F_g(A)$ as a cohomology class before calling it free energy.
- 472. Define λ_g^{FP} .
- 473. Define ω_g .
- 474. Prove $(d_{\text{fib}}^{\text{ord}})^2 = r_A(z) \cdot \omega_g$.
- 475. State whether this is an equality of operators, cohomology classes, or curvature forms.
- 476. Define scalar projection tr_{diag} .
- 477. Prove scalar projection gives $\kappa(A)\lambda_g$.
- 478. Define uniform-weight lane precisely.
- 479. Prove cross-channel term vanishes on uniform-weight lane.
- 480. Define cross-channel correction $\delta F_g^{\text{cross}}$ as mixed-edge stable-graph sum.
- 481. Prove no cross-channel graph at genus one.
- 482. Prove mixed-edge graphs exist for $g \geq 2$.
- 483. Compute genus-one Heisenberg curvature.
- 484. Compute genus-one affine curvature.
- 485. Compute genus-one Virasoro curvature.
- 486. Compute genus-two first cross-channel term for W_3 .
- 487. Define stable graph weights.
- 488. Define OPE channel labels on edges.
- 489. Prove graph sum converges in each finite window.
- 490. Prove strict ML passage to all weights.
- 491. Define modular operad action on the genus tower.
- 492. Prove separating clutching compatibility.
- 493. Prove non-separating clutching compatibility.
- 494. Prove genus-raising $\hbar\Delta$ satisfies BV identity.
- 495. State if the modular bar is wheeled, modular, or cyclic.
- 496. Define orientation of loop edge.

497. Compute loop sign.
498. Define Hodge class normalisation.
499. Compare with Faber–Pandharipande only after defining map.
500. Remove all physical “free energy” language from theorem statements unless class is analytic.
501. Define analytic partition function only for VOAs satisfying convergence.
502. Separate formal genus series from analytic free energy.
503. Define holomorphic anomaly term separately.
504. State E_2 quasi-modularity at genus one.
505. Prove ordered-to-symmetric non-splitting at genus one.
506. Identify obstruction with $E_2 - E_2$.
507. Define non-holomorphic completion.
508. Prove no holomorphic section exists.
509. State relation to KZB connection.
510. Define KZB connection for elliptic curve.
511. Prove flatness under OPE/Jacobi conditions.
512. Define higher-genus prime-form propagator.
513. Prove its local diagonal residue is one.
514. Compute its period ambiguity.
515. Relate period ambiguity to curvature.
516. Define genus-two KZB data.
517. State Fay trisecant identity precisely.
518. Prove DDYBE only on face-model scope.
519. Separate KZB flatness from Yangian duality.
520. Define modular characteristic κ from genus-one curvature.
521. Prove same κ equals degree-two averaged residue.
522. Prove naturality of κ under Koszul duality where defined.
523. Define complementarity K_κ only after both $\kappa(A)$ and $\kappa(A^\dagger)$ exist.
524. Compute K_κ for G.
525. Compute K_κ for L.
526. Compute K_κ for C.
527. Compute K_κ for M.
528. State B/K3 row as cross-volume bucket, not Vol I row.
529. Prove $K_\kappa = 8$ for Mukai witness only under source hypotheses.
530. Do not call 8 universal.
531. Define κ_{BKM} separately.
532. Define κ_{fiber} separately.
533. Define κ_{cat} separately.
534. Add theorem: K3 constants live on five lanes.
535. Prove no transfer to exceptional Yangian RTT without comparison maps.
536. Add a normalization table for $\lambda_g, \kappa, K_\kappa$.
537. Add exact signs for genus-raising operator.
538. Prove $\hbar$ -degree assignments.
539. Define all formal variables $t, q, \hbar, z, \omega$.
540. Remove every use of a variable before definition.

H. Theorem C, derived centre, symplectic complementarity: 541–600

541. Define $Z_{\text{ch}}^{\text{der}}(A) = R\text{Hom}_{A-A}^{\text{fact}}(A, A)$.
542. Define its brace dg algebra structure.
543. Define its E_2 -structure only under chiral Deligne hypotheses.
544. Define $\text{SC}_{\text{ch}, \text{top}}$ output only on the pair $(Z_{\text{ch}}^{\text{der}}(A), A)$.
545. State $B(A)$ is not $\text{SC}_{\text{ch}, \text{top}}$.
546. Prove $B(A)$ computes Z^{der} via diagonal resolution.
547. Define open/closed comparison map OCA.
548. Do not identify Z^{der} with physical bulk without OCA.
549. Define complementarity spaces $Q_g(A)$.
550. Define $Q_g(A^\dagger)$.

551. Define ambient $H^*(M_g, Z(A))$.
552. Prove C0/C1 linear complementarity.
553. State C2 shifted-symplectic enhancement separately.
554. Define deformation stack $\bar{M}_A$.
555. Define $\bar{M}_{A^!}$.
556. Define $\bar{M}_{\text{comp}}$.
557. Define tangent complexes.
558. Prove perfectness.
559. Prove nondegeneracy of pairing.
560. Prove finiteness of modular envelope.
561. Define isotropic embeddings $i_A, i_{A^!}$.
562. Prove isotropicity.
563. Prove Lagrangian maximality under complementarity.
564. Define derived intersection $\bar{M}_A \times_{\bar{M}_{\text{comp}}}^h \bar{M}_{A^!}$.
565. Prove it is represented by twisted tensor product $K_\tau(A, A^!)$.
566. State this representation is the missing converse if not proved.
567. Prove acyclicity of K_τ implies Koszulness.
568. Prove transversality implies acyclicity.
569. Do not state Lagrangian characterization unconditionally.
570. Define PTVV hypotheses explicitly.
571. State shifted symplectic degree -1 .
572. Check signs of symplectic pairing.
573. Prove cyclic pairing is closed.
574. Prove differential is skew-adjoint.
575. Prove compatibility with modular bracket.
576. Define BV package if C2 uses BV.
577. Separate classical BV from quantum BV.
578. State QME not proved unless analytic SDR exists.
579. Define self-image renormalisation if used.
580. Define image-choice Stokes data if used.
581. Keep Khan–Zeng bridge classical unless all-loop proof supplied.
582. Define derived centre of each E_1 presentation separately.
583. Do not assume the five derived centres coincide.
584. State comparison maps among them if any.
585. Prove their equivalence on Koszul generic locus only under H1–H4.
586. Define categorical Drinfeld centre separately from chiral derived centre.
587. Prove any comparison between them.
588. Separate E_2 centre theorem from E_3 topologization.
589. Define conformal vector needed for E_3 .
590. Prove BRST exactness of translations.
591. State critical level failure of topologization.
592. Compute affine $\mathfrak{sl}_2$ derived centre explicitly.
593. Compute Heisenberg derived centre explicitly.
594. Compute Virasoro derived centre under generic hypotheses.
595. Compute $\bar{W}_3$ derived centre under generic hypotheses.
596. State centre computations as checks, not definitions.
597. Remove "bulk" from scalar complementarity theorem.
598. Define scalar trace map from Z^{der} to κ .
599. Prove K_κ is trace shadow of C1 plus Theorem D.
600. Make C theorem a three-level theorem: C0 algebraic, C1 derived-centre, C2 shifted-symplectic.

I. Standard family computations: 601–700

601. For Heisenberg, define generator $a(z)$ and OPE.
602. Compute $B(H_k)$ through bar degree three.
603. Prove Koszulness of H_k .
604. Compute $H_k^!$.

605. Compute $r_{H_k}(z) = k/z$.
606. Compute $\kappa(H_k) = k$.
607. Compute $CH_{\text{ch}}^{\bullet}(H_k)$.
608. Check $K_{\kappa}(H_k, H_k^1) = 0$.
609. For affine $V_k(\mathfrak{g})$, define OPE with normalized κ_{ab} .
610. State $k \neq -h^{\vee}$.
611. Compute bar differential in degree two.
612. Compute cubic shadow $\kappa_{\mathfrak{g}}(x, [y, z])$.
613. Prove quartic shadow vanishes by Jacobi.
614. Prove $r_{\max} = 3$.
615. Compute r_{coll} .
616. Compute r_{KZ} .
617. Prove level reflection.
618. Compute $\kappa(V_k(\mathfrak{g}))$ with fixed normalization.
619. Compute K_{κ} for the level-reflected pair.
620. State critical level separately.
621. Define Feigin–Frenkel center separately.
622. Do not treat critical centre as generic specialization.
623. For $\beta\gamma$, define fields and weights.
624. Compute bar complex degree two.
625. Compute degree three.
626. Prove quartic birth if claimed.
627. Define weight-changing line.
628. Prove $m_3 = 0$ on that line if claimed.
629. Compute $\kappa(\beta\gamma)$.
630. Identify Koszul dual only with correct parity.
631. Separate $\bar{b}c$ and free fermion conventions.
632. State symplectic boson specialization $\lambda = 1/2$.
633. Prove equivariant bar statements at $\lambda = 1/2$.
634. For Virasoro, define OPE.
635. Define stress tensor normalization.
636. Compute bar degree two.
637. Compute cubic gravitational shadow.
638. Compute quartic contact coefficient.
639. Prove $S_4(\text{Vir}_c) \neq 0$ at generic c .
640. State exceptional central charges where formula degenerates.
641. Fix $c = 0, c = -22/5, c = 26, c = 13$ statuses.
642. Compute $\kappa(\text{Vir}_c) = c/2$.
643. Compute $A^1 = \text{Vir}_{26-c}$ only under theorem.
644. Check $K_{\kappa} = 13$.
645. Do not call Virasoro self-dual except at $c = 13$.
646. For W_3 , define generators T, W .
647. State OPE $W(z)W(w)$ fully.
648. Define composite field $\Lambda =: TT : -\frac{3}{10}\partial^2 T$.
649. Check coefficient $\frac{32}{22+5c}$.
650. Compute quartic shadow from Λ .
651. Compute $W_3 \kappa$.
652. Compute $K_{\kappa}(W_3, W_3^1) = 250/3$ if claimed.
653. Define W_N spin set.
654. Compute $H_N - 1$ contribution to $\kappa(W_N)$.
655. Prove $K_c(W_N) = 4N^3 - 2N - 2$.
656. Prove $K_{\kappa}(W_N) = (H_N - 1)(4N^3 - 2N - 2)$.
657. Separate K_c ghost spin-sum from K_{κ} .
658. For Bershadsky–Polyakov, define fields.
659. State central charge.
660. Compute its shadow lines separately.

- 661. Do not include BP in principal W_N class without DS orbit statement.
- 662. Define subregular DS reduction.
- 663. Prove PBW–Slodowy collapse.
- 664. Prove Appell formula if used.
- 665. State unbounded canonical degree theorem precisely.
- 666. For $N = 2$ SCA, define OPEs.
- 667. Compute multi-channel shadow.
- 668. Define spectral flow action.
- 669. Prove Koszulness if claimed.
- 670. Otherwise mark Koszulness as conditional.
- 671. For lattice VOAs, define even lattice L .
- 672. Define 2-cocycle.
- 673. Prove locality iff cocycle symmetry condition.
- 674. Define ordered E_1 deformation for nonsymmetric cocycle.
- 675. Compute A_2 quantum lattice example.
- 676. State root-of-unity specializations.
- 677. For moonshine, distinguish Monster VOA from K3/Mukai lane.
- 678. Do not use moonshine constants as K3 constants without comparison.
- 679. For symmetric orbifolds, define twist sectors.
- 680. State fixed-point Koszulness hypotheses.
- 681. State DMVV formula as character identity, not bar theorem.
- 682. For triplet $W(p)$, state C_2 -cofiniteness.
- 683. State non-free generation.
- 684. State Koszulness conjecture.
- 685. Define high-degree Hochschild defect.
- 686. For admissible levels, define simple quotient.
- 687. Prove quotient bar comparison.
- 688. State generic universal affine theorem does not automatically descend.
- 689. For super-Yangians, define parity conventions.
- 690. Prove Berezinian shifts.
- 691. State sign changes in bar differential.
- 692. Define supertrace normalization.
- 693. For exceptional Yangians, define finite RTT obstruction.
- 694. Prove type-by-type finite window results.
- 695. Do not extrapolate A_n RTT to exceptional types.
- 696. For Kac–Moody exceptional E_6, E_7, E_8 , compute shadow data directly.
- 697. Add table with theorem status: proved, conditional, conjectural.
- 698. Add table with ambient: raw, finite, completed.
- 699. Add table with object: $A, B, A^i, A^!, Z^{der}$.
- 700. Delete or quarantine any example lacking all three tables.

J. H_{Δ_5} , K3, Hall, BKM, CY comparison: 701–780

- 701. Define H_{Δ_5} only after D1–D5.
- 702. Before D1–D5, call it $H_{\Delta_5}^{\text{target}}$.
- 703. Define compact Hall source.
- 704. Name the K3 source.
- 705. Define Hall algebra in finite height/charge/weight window.
- 706. Define associated graded Hall algebra.
- 707. Prove identification with finite Borcherds truncation.
- 708. Define $\mathfrak{g}_{\Delta_5}$.
- 709. State root multiplicities.
- 710. State Euler form.
- 711. Prove both match Borcherds product for Δ_5 .
- 712. Define augmentation.
- 713. Define coproduct.
- 714. Define completion.

715. Define $\hbar^2 = -1/8$ specialization.
716. Prove specialization respects truncations.
717. Define real-root set.
718. Define imaginary simple roots.
719. Define Weyl vector.
720. Define super-parity signs.
721. Prove parity transported through Hall, BKM, and bar.
722. Prove PBW spectral sequence degenerates in each finite window.
723. Prove no extra root spaces.
724. Prove no hidden central summands.
725. Prove no torsion classes.
726. Define comparison maps from $B_{ch}^{E_1}(H_{\Delta_5})$ to Hall truncations.
727. Define comparison maps from cobar to Hall truncations.
728. Prove maps preserve product.
729. Prove maps preserve coproduct.
730. Prove maps preserve differential.
731. Prove maps preserve augmentation.
732. Prove maps preserve inverse-system maps.
733. Prove maps preserve averaging.
734. Prove finite-window counits are quasi-isomorphisms.
735. Prove ML passage to limit.
736. Only then state $\Omega B(H_{\Delta_5}) \simeq H_{\Delta_5}$.
737. Define K3 Mukai scalar.
738. Prove $K_\kappa = 8$ under comparison.
739. Distinguish $K_\kappa = 8$ from $\kappa_{\text{BKM}} = 5$.
740. Distinguish both from $\kappa_{\text{fiber}} = 24$.
741. Distinguish all from $\kappa_{\text{cat}} = 0$.
742. Define $\kappa_{\text{Heis}} = 3$ for $K3 \times E$ lane.
743. Prove lane separation theorem.
744. State no number transfers between lanes without comparison.
745. Define two-stage CY-to-chiral functor $\Phi_d^{(\Sigma, C)}$.
746. Define Φ_d^{FA} .
747. Define $Sp_{\Sigma, C}^{ch}$.
748. Prove their composition on the stated surface.
749. Do not write one-stage Φ_d where two-stage is meant.
750. Define reference curve.
751. Define transverse manifold Σ_{d-1} .
752. Prove factorization-homology hypotheses.
753. Define $HH^{-1} = 0$ slice if used.
754. State $d = 2$ slice separately from $d \geq 3$.
755. Separate Hilbert-scheme Fock bridge from categorical lift.
756. Mark categorical lift conjectural unless proved.
757. Define CoHA source.
758. Define Hall–Drinfeld double.
759. Prove Drinfeld double structure.
760. Define Lusztig specialization.
761. Prove finite quantum group truncation.
762. State root-of-unity order.
763. Compute PBW dimensions for ζ_8 if claimed.
764. Compute PBW dimensions for ζ_{12}, ζ_{24} if claimed.
765. Prove conditional PBW dimensions or mark conditional.
766. Define Humbert divisors.
767. Define admissible Heegner union.
768. Prove bi-unipotent Malcev ladder.
769. State codim-3 transversality theorem with proof.
770. State codim-4 emptiness theorem with proof.

- 771. Define nearby-cycle functor ψ_n^w .
- 772. Prove bar filtration compatibility with nearby cycles.
- 773. Prove $\lim^1$ obstruction formula.
- 774. Define Fourier coefficient of $\phi_{-2,1}$ used.
- 775. Prove Baily–Borel compatibility.
- 776. Define Heegner comparison gates.
- 777. State H_{Δ_5} theorem as recognition, not construction, until 701–776 are done.
- 778. Move speculative associator $\Phi^{\text{Sieg–Bor}}$ to conjecture unless constructed.
- 779. Prove pentagon through all orders or state order- $\hbar^3$ only.
- 780. Do not call scalar target data an associator on Hall realization.

K. Physics and open/closed bridges: 781–850

- 781. Define physical bulk observables $O_{\text{bulk}}^{\text{phys}}(A)$.
- 782. Define open/closed comparison map OCA.
- 783. Prove OCA is a quasi-isomorphism before saying “holographic”.
- 784. Define $\text{SC}_{\text{ch},\text{top}}$ operation spaces.
- 785. Give minimal generators only if fully known.
- 786. State codimension-one generators.
- 787. State codimension-two relations.
- 788. Prove Arnold relation in closed colour.
- 789. Prove Stasheff relation in open colour.
- 790. Prove mixed square relation.
- 791. Define no-open-to-closed rule.
- 792. Prove closed colour is operadic retract.
- 793. Prove derived centre carries $\text{SC}_{\text{ch},\text{top}}$ structure.
- 794. Prove bar complex does not carry $\text{SC}_{\text{ch},\text{top}}$.
- 795. Define E_3 topologization.
- 796. State conformal vector hypothesis.
- 797. State non-critical level hypothesis.
- 798. Prove holomorphic translations are BRST-exact.
- 799. Prove $E_2 \rightarrow E_3$ upgrade on BRST cohomology.
- 800. State critical level failure.
- 801. Define BV action for finite-type PVA.
- 802. Prove CME iff PVA Hamiltonian condition.
- 803. Do not state QME unless renormalization package is built.
- 804. Define analytic SDR.
- 805. Define Stokes image choice.
- 806. Define self-image renormalisation.
- 807. Define field-parity signs.
- 808. Prove all-loop QME only after 804–807.
- 809. Define boundary vertex algebra from BV data.
- 810. Prove boundary OPEs.
- 811. Define reflected weight identity.
- 812. Prove it or mark conditional.
- 813. Define Chern–Simons comparison for affine Kac–Moody.
- 814. Prove abelian Chern–Simons yields Heisenberg centre.
- 815. Prove nonabelian HT Chern–Simons yields affine currents.
- 816. State perturbative finiteness hypotheses.
- 817. Define Costello–Gwilliam factorization algebra used.
- 818. State whether theory is classical or quantum.
- 819. Separate 3d HT bulk from curve-level chiral algebra.
- 820. Define slab reduction if used.
- 821. Prove slab reduction for $\mathcal{W}_3$ PVA.
- 822. Define Poisson sigma model.
- 823. Prove HT enhancement.
- 824. Define AGT comparison only as external motivation unless theorem.

- 825. Define line-operator category.
- 826. Prove evaluation modules generate it if claimed.
- 827. State thick-generation hypotheses.
- 828. Prove MC3 on evaluation-generated core.
- 829. Mark full DK boundary open.
- 830. Define Baxter retraction as type-A only.
- 831. Do not apply Baxter retraction outside type A.
- 832. Define quantum group parameter q .
- 833. State q -convention bridge.
- 834. Check KZ monodromy signs.
- 835. Check modular T -matrix conventions.
- 836. Check self-monodromy vs mutual monodromy.
- 837. Define framing square root.
- 838. Prove all convention conversions.
- 839. Remove physics metaphors from theorem names.
- 840. Use physics as corollaries only after algebraic theorem.
- 841. Define entropy formulas only as character asymptotics.
- 842. Prove BTZ entropy only under modular-invariance and large- N conditions.
- 843. Define celestial comparison if kept.
- 844. Prove factorization comparison.
- 845. Define celestial algebra input.
- 846. State scope of celestial theorem.
- 847. Remove unsupported AdS/CFT identifications.
- 848. Replace "holographic datum" by a seven-object tuple with maps.
- 849. Prove reconstruction from tuple.
- 850. State failure modes when maps are absent.

L. Arithmetic and modular forms: 851–900

- 851. Define the arithmetic shadow tower as a functor from Θ_A .
- 852. Define moment L -functions precisely.
- 853. State Rankin–Selberg integral with convergence domain.
- 854. Prove meromorphic continuation where claimed.
- 855. Define Hecke action on $M_{1,1}$.
- 856. Define lift of Hecke action to $\mathfrak{g}_A^{mod}$.
- 857. State prime-locality conjecture precisely.
- 858. Prove it for lattice VOAs if claimed.
- 859. State non-lattice prime-locality as hypothesis.
- 860. Define CPS hypotheses.
- 861. Define automorphic representation π_r .
- 862. State functoriality $GL(2) \rightarrow GL(r+1)$ as hypothesis for $r \geq 5$.
- 863. Do not claim Ramanujan beyond known functoriality unconditionally.
- 864. State Kim–Sarnak bound where unconditional.
- 865. Define Hecke–Newton closure.
- 866. Prove it for lattice finite Hecke span.
- 867. Define non-lattice extension separately.
- 868. Define irrational extension separately.
- 869. Mark irrational extension conditional.
- 870. Define Dirichlet L -functions arising from shadow metric.
- 871. Prove spectral decomposition.
- 872. Define Roelcke–Selberg basis.
- 873. Prove compatibility with chiral partition function.
- 874. Define sewing operator.
- 875. Prove sewing–Selberg formula.
- 876. Define Saito–Kurokawa lift.
- 877. Prove genus-two bridge only where known.
- 878. Define Böcherer bridge.

- 879. State critical-line access as conditional if not proved.
- 880. Define Borchers product Δ_5 .
- 881. Compute $c_{\Delta_5}(0)$.
- 882. Prove $\kappa_{\text{BKM}} = c(0)/2$.
- 883. State CHL scope for $N \in \{1, 2, 3, 4, 6\}$.
- 884. Do not transfer CHL constants to K3 Mukai K_κ .
- 885. Define Igusa cusp form Φ_{10} .
- 886. Prove any appearance in associator cocycle.
- 887. Define η^{24} quotient if used.
- 888. State modular weight.
- 889. Prove transformation law.
- 890. Define Fricke LDP.
- 891. Prove sub-leading correction at each node.
- 892. Define Shimura–Waldspurger conversion.
- 893. State weights $k + 1 \in \{7, 9, 11\}$.
- 894. Prove conversion or cite theorem precisely.
- 895. Define Yetter–Drinfeld Schauenburg bracket.
- 896. Compute $\delta^{(n)}$ for claimed n .
- 897. Do not mix arithmetic brackets with bar differential without map.
- 898. Define GRT action on K3-BKM if used.
- 899. Prove transitivity or state conjectural.
- 900. Move arithmetic beyond theorem spine unless it proves a needed algebraic claim.

M. Exposition-as-mathematics repairs: 901–1000

- 901. Replace “Platonic” in theorem names by exact mathematical adjectives.
- 902. Keep “Platonic” only in informal remarks.
- 903. Put all conjectures in numbered conjecture environments.
- 904. Put all conditional theorems in theorem environments with hypotheses first.
- 905. No theorem title should contain “conditional” unless its hypotheses are also complete.
- 906. Every theorem should say raw/finite/completed.
- 907. Every theorem should say ordered/symmetric.
- 908. Every theorem should say chain/derived/coderived.
- 909. Every theorem should say $E_1/E_\infty/\text{SC}/E_3$.
- 910. Every theorem should state curve genus scope.
- 911. Every theorem should state critical-level exclusions.
- 912. Every theorem should state genericity exclusions.
- 913. Every theorem should state finite-type hypotheses.
- 914. Every theorem should state completion hypotheses.
- 915. Every theorem should state dualizability hypotheses.
- 916. Every theorem should state holonomicity hypotheses.
- 917. Every theorem should state base-change hypotheses.
- 918. Every proof should begin by naming the ambient category.
- 919. Every proof should identify the differential.
- 920. Every proof should identify the filtration.
- 921. Every proof should identify the spectral sequence.
- 922. Every proof using convergence should cite the convergence theorem.
- 923. Every proof using inverse limits should kill $\lim^1$.
- 924. Every proof using duality should check finite/continuous duality.
- 925. Every proof using Verdier duality should check holonomicity.
- 926. Every proof using Gysin maps should check orientation.
- 927. Every proof using pushforward should check properness.
- 928. Every proof using quotient by Σ_n should check equivariance.
- 929. Every proof using $1/n!$ should state characteristic zero.
- 930. Every proof using R -matrix descent should state R -twisted equivariance.
- 931. Every proof using PBW should state associated graded.
- 932. Every proof using Koszulness should state diagonal concentration.

933. Every proof using formality should state model and obstruction groups.
934. Every proof using A_∞ transfer should state SDR data.
935. Every proof using modular operad should state graph convention.
936. Every proof using stable graphs should define legs, edges, genus.
937. Every proof using signs should cite determinant-line convention.
938. Every proof using ghosts should define BRST complex.
939. Every proof using central charge should define convention.
940. Every proof using κ should define trace map.
941. Every proof using K_κ should prove $A^!$ exists.
942. Every proof using K_c should prove central charge of $A^!$.
943. Every proof using K_{ghost} should state presentation dependence.
944. Every proof using OPE tables should provide full coefficients.
945. Every OPE coefficient should match one normalization appendix.
946. Every table entry should cite theorem/proposition proving it.
947. Every "proved" table status should be backed by a proof in the manuscript.
948. Every "conditional" table status should list the exact conditions.
949. Every "open" status should name the obstruction complex.
950. Remove duplicate theorem statements that differ only rhetorically.
951. Merge repeated sign appendices into one sign theorem.
952. Merge repeated "five-object firewall" remarks into one early proposition.
953. Cross-reference the firewall everywhere.
954. Move long motivational physics paragraphs after theorem proof.
955. Move all unproved cross-volume claims to "Comparison conjectures."
956. Move all K3 recognition targets to one conditional chapter.
957. Move all arithmetic consequences after algebraic core.
958. Move all examples after main theorem.
959. Put Heisenberg as first verification theorem.
960. Put affine Kac–Moody as second verification theorem.
961. Put $\beta\gamma$ as third verification theorem.
962. Put Virasoro/ $\widehat{W}_3$ as class-M completion test.
963. Put H_{Δ_5} after class-M completion, not before.
964. Add dependency graph with arrows only for proved implications.
965. Remove arrows for conjectural comparisons.
966. Replace "therefore" by "under hypotheses" whenever a comparison theorem is used.
967. Replace "universal" by "natural in the stated category" unless universal property is proved.
968. Replace "canonical" by "specified" unless uniqueness is proved.
969. If canonical, prove contractibility of choices.
970. Define all "chosen vacuum" data.
971. Prove independence of vacuum if claimed.
972. Define open factorization category at level 0.
973. Prove Morita equivalence to module category if used.
974. Separate Morita equivalence from algebra isomorphism.
975. Define level alphabet once.
976. Prove level signatures are exhaustive only for Open quadrant.
977. Do not use level signatures to transfer CY claims.
978. Define quadrant grid.
979. Prove cross-quadrant maps preserve structures.
980. State when cross-quadrant maps are conjectural.
981. Add notation table before main theorem.
982. Remove symbols unused after definition.
983. Avoid reusing C for curve, coalgebra, and category in same proof.
984. Avoid reusing D for divisor, differential, and duality in same proof.
985. Use $\mathbb{D}$ only for Verdier duality.
986. Use d only for differentials.
987. Use Δ separately for diagonal, coproduct, and discriminant with subscripts.
988. Use $FM, FM^{\log}, FM^{ord}$ consistently.

- 989. Use B^{ord} , B^{Σ} , $B^{mod,log}$ consistently.
- 990. Use CH_{ch} , HH_{alg} , THH consistently.
- 991. Use Z_{ch}^{der} , $Z(A)$, Z_{Dr} distinctly.
- 992. Add an index of hypotheses H1–H4, CP1–CP3, D1–D5, FH, OCA.
- 993. Never reference a hypothesis package before definition.
- 994. Never cite “§??”; replace all missing cross-references.
- 995. Replace all “standard landscape” references by a defined finite list.
- 996. Prove landscape coverage or call it an atlas, not classification.
- 997. State the exact arXiv abstract from the theorem spine only.
- 998. Put conjectural physics/arithmetic in the final paragraph of the abstract only if labelled.
- 999. Make the first 20 pages prove the core theorem’s definitions are type-correct.
- 1000. Make the last theorem of the paper a single commutative diagram showing exactly which invariants survive

$$B^{ord}(A) \longrightarrow B^{\Sigma}(A)$$

and exactly which do not.

The shortest route to the strongest paper

The paper’s inner music is not “everything is one thing.” Its inner music is:

collision geometry $\Rightarrow$ ordered bar coalgebra $\Rightarrow$ MC element $\Rightarrow$ averaged scalar invariants

with Verdier duality, derived centre, and physics comparison living on **separate but functorially related** branches.

The strongest publishable version is a paper where the main theorem is not a slogan but a typed equivalence of completed homotopical objects:

signed log-FM cooperad + PBW-Koszul chiral algebra + strict ML completion $\implies$ bar-cobar inversion, Verdier

and every K3, Yangian, W -algebra, $SC_{ch,top}$, or arithmetic statement is either a proved functorial image of that theorem or is explicitly labelled as a conjectural comparison surface.

       Sources